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EXPERIMENT-3

Aim : To Perform various GIT operations on local and Remote repositories using GIT Cheat-Sheet

Pull and Push Processes

The pull command used to fetch the repository from github to local while push is used to commit files from local repository to Github.

Push☐ Push changes to Web repository Pull ☐

Pull changes to Local repository

The following commands are used for pull and push repositories

A) Push command

```
PS C:\Users\HP 15L\Desktop\T21-75> git init
Initialized empty Git repository in C:/Users/HP 15L/Desktop/T21-75/.git/
PS C:\Users\HP 15L\Desktop\T21-75>
```

```
$ git remote add origin https://github.com/bhushanjadhav1/siesworkshop.git
```

```
Initialized empty Git repository in C:/Users/HP 15L/Desktop/
PS C:\Users\HP 15L\Desktop\T21-75 git remote add origin https:ryga:
tical-star/SEPM_PRAC.git
PS C:\Users\HP 15L\Desktop\T21-75>
```

$ git remote show origin

```
PS C:\Users\HP 15L\Desktop\T21-79> git remote show origin
* remote origin
  Fetch URL: https://github.com/tsecpractical-star/SEPM_PRAC.git
  Push  URL: https://github.com/tsecpractical-star/SEPM_PRAC.git
  HEAD branch: main
  Remote branch:
    main new (next fetch will store in remotes/origin)
PS C:\Users\HP 15L\Desktop\T21-79>
```

```
PS C:\Users\HP 15L\Desktop\T21-79> git commit -m "initial commit"
[master (root-commit) 0b77317] initial commit
 1 file changed, 10 insertions(+)
 create mode 100644 exp5.py
PS C:\Users\HP 15L\Desktop\T21-79> git push -u origin main
error: src refspec main does not match any
```

If you add remote again then will show you fatal error.

$ git remote add origin https://github.com/bhushanjadhav1/Myrepository.git

fatal: remote origin already exists. So, to delete origin rm origin

command is used

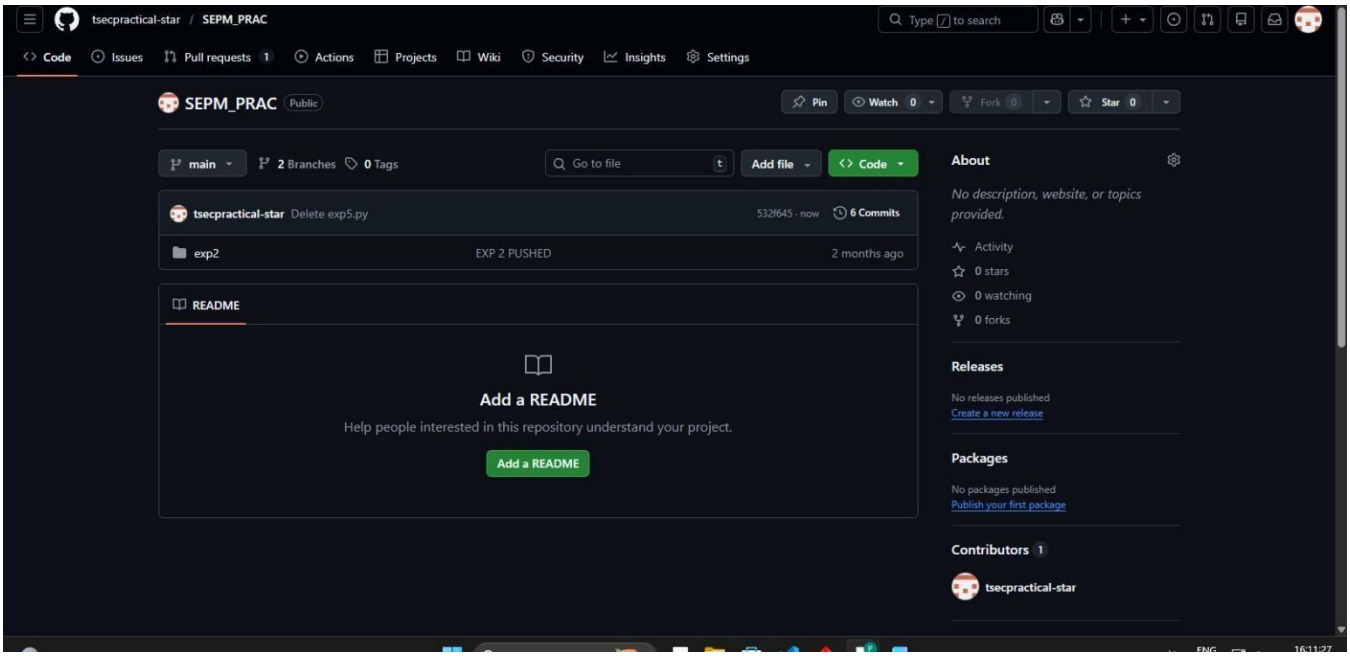
$ git remote rm origin

Now, to push the local repository to remote github following command is used $

git push -u origin master

```
PS C:\Users\HP 15L\Desktop\T21-79> git branch -m Main
PS C:\Users\HP 15L\Desktop\T21-79> git push -u origin Main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 24 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 657 bytes | 657.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Main' on GitHub by visiting:
remote:      https://github.com/tsecpractical-star/SEPM_PRAC/pull/new/Main
remote:
To https://github.com/tsecpractical-star/SEPM_PRAC.git
 * [new branch]      Main -> Main
branch 'Main' set up to track 'origin/Main'.
PS C:\Users\HP 15L\Desktop\T21-79>
```

Now you can check the github for updated contents.



B) Pull Changes

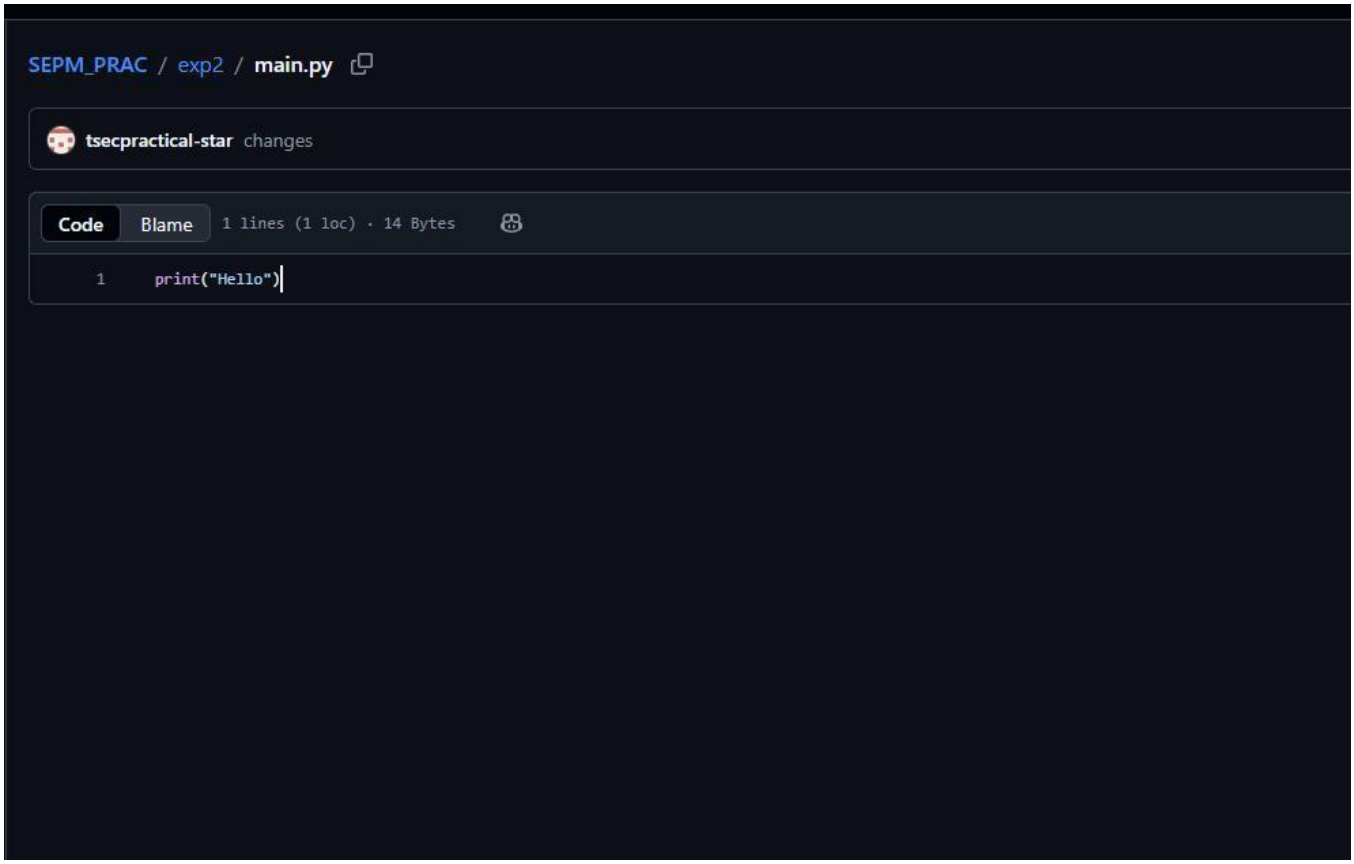
Pull command is used to download the remote updated repository into local one. The command for download is

```
$ git pull
```

```
PS C:\Users\HP 15L\Desktop\T21-79> git pull
remote: Enumerating objects: 2, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 1.66 KiB | 113.00 KiB/s, done.
From https://github.com/tsecpractical-star/SEPM_PRAC
 * [new branch]      main       -> origin/main
Already up to date.
PS C:\Users\HP 15L\Desktop\T21-79>
```

C) Fetch

Suppose you have a file in github and you have changes that.



ORIGINAL FILE

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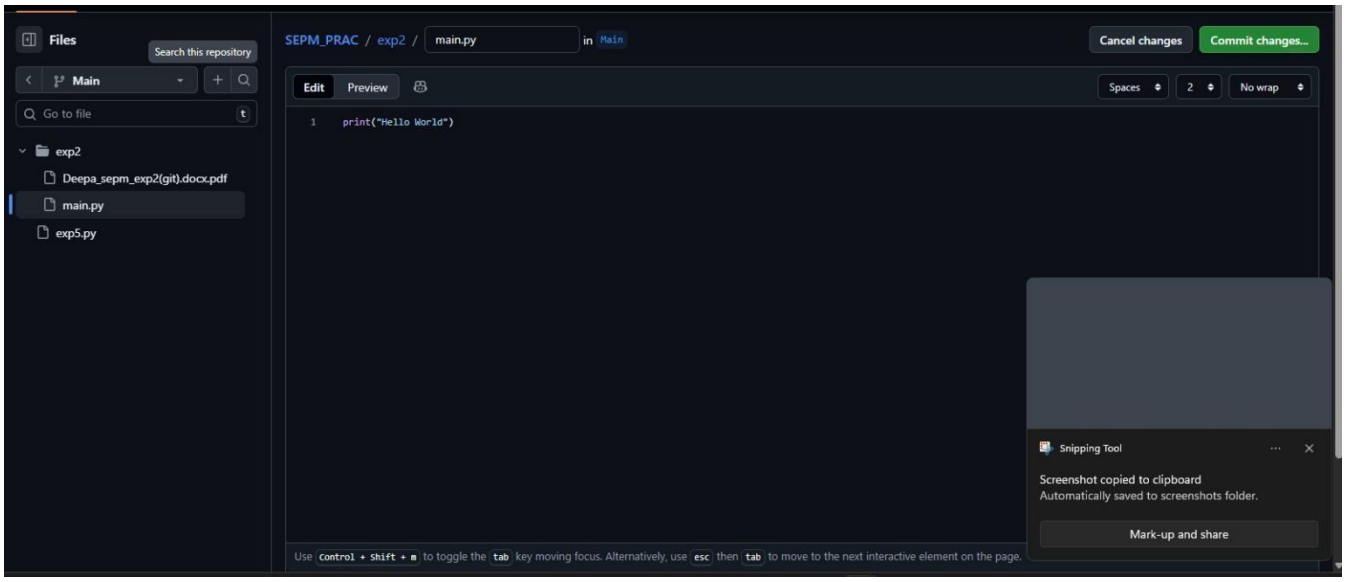

Changed File

Now we use fetch command to fetch the changes, which will show you both the files like original and changed in local repository.

$ git fetch

```
PS C:\Users\HP 15L\Desktop\T21-79> git fetch
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.61 KiB | 178.00 KiB/s, done.
From https://github.com/tsecpractical-star/SEPM_PRAC
   011b9ae..09bf1dd  Main       -> origin/Main
 * [new branch]      main       -> origin/main
PS C:\Users\HP 15L\Desktop\T21-79>
```

Here fetch will not show you like updated changes file as like push. So use merge command to merge the changes so use following command for merge. $ git merge origin/master

```
PS C:\Users\HP 15L\Desktop\T21-79> git merge origin/Main
>>
Merge made by the 'ort' strategy.
 exp5.py | 10 ----------
 1 file changed, 10 deletions(-)
 delete mode 100644 exp5.py
PS C:\Users\HP 15L\Desktop\T21-79>
Main  0↓ 2↑  ⊗ 0 ⚠ 0  0  Ln 1, Col 15  Spaces: 4  UTF-8
```

Online Resource: https://dzone.com/articles/top-20-git-commands-with-examples